HW 9

1

Analyze circuit P1 shown below. Determine the transition table, flow table, and output table. For the flow table let A denote the state at which $(y_1 y_2) = 00, B = 01, C = 11, D = 10$.

Answer

Where $z = x_2 y_1 y_2'$

$S_1 = x_2 y_2$
$R_1 = x_1 y_2$
$S_2 = x_2 y_1' + x_1 y_1$
$R_2 = x_1 + x_2$

$S_1 R_1 S_2 R_2 z$ Value Table

$y_1 y_2 \backslash x_1 x_2$	00	01	11	10
00	00000	00110	00110	00010
01	00000	10110	11110	01010
11	00000	10010	11110	01110
10	00000	00011	00111	00110

Transition Table

$y_1 y_2 \backslash x_1 x_2$	00	01	11	10
A	(A)	B	B	(A)
B	(B)	D	C	A
C	(C)	D	(C)	B
D	(D)	(D)	C	C

Flow Table

$y_1 y_2 \backslash x_1 x_2$	00	01	11	10
A	(A)	D	C	(A)
B	(B)	D	C	A
C	(C)	D	(C)	A
D	(D)	(D)	C	A

2

You are to analyze the circuit P2:

a

Insert fictional amplifiers to break all feedback loops. Use as few amplifiers as possible.

Answer

I would insert these amplifiers at the output of each NOR gate, which feeds backwards and into Q and P.

b

Write excitation functions.

Answer

$L = (x_1 + Q + P + x_3)'$

$Q_{next} = (x_1 + x_2 + L + Q)'$

$P_{next} = (P + L + x_2 + x_3)'$

c

Form the flow table.

Answer

Next value for LQP

$QP \backslash x_1x_2x_3$	000	001	010	011	100	101	110	111
A 00	100	010	100	000	001	000	000	000
B 01	010	010	000	000	000	000	000	000
C 11	000	000	000	000	000	000	000	000
D 10	001	000	000	000	001	000	000	000

Transition value

$QP \backslash x_1x_2x_3$	000	001	010	011	100	101	110	111
A 00	(A)	D	(A)	(A)	B	(A)	(A)	(A)
B 01	D	D	A	A	A	A	A	A
C 11	A	A	A	A	A	A	A	A
D 10	B	A	A	A	B	A	A	A

Flow State

$QP \backslash x_1x_2x_3$	000	001	010	011	100	101	110	111
A 00	(A)	D!	(A)	(A)	B!	(A)	(A)	(A)
B 01	D!	D!	A	A	A!	A	A	A
C 11	A	A!	A	A	A!	A	A	A
D 10	B!	A!	A	A	B!	A	A	A

States with an `!` do not converge.

d

Convert the circuit to one constructed of NAND gates with the same behavior. Use as few additional gates as possible.

✓ **Answer**

Invert the inputs and invert the outputs and they will be structurally the same.

3

A suspicious person claims that circuit P3 is a JK flip-flop without a master-slave configuration. Suppose the delay of each gate is τ, and that inputs change only at times that are integer multiples of τ.

a

Is this circuit a latch or a flip-flop? Explain your reasoning.

✓ **Answer**

When clock is 0, Q will stay stable no matter how J and K change. But when clock is 1, J transparently sets Q and K transparently unsets Q, thus it is a latch.

b

Give a timing diagram for the circuit, starting from state $(C, J, K, Q) = (0, 1, 0, 0)$, when C changes to 1.

What Is the minimum time for which C must remain 1 for the circuit to enter the stable state $(C, J, K, Q) = (1, 1, 0, 1)$?

✓ **Answer**

Time $\times\tau$	Action
0	$C \to 1$
1	$NAND_1 \to 0$
2	$Q \to 1$
3	$\bar{Q} \to 0$, $NAND_1 \to 1$

Clock must be high for at least 2τ

c

Give a timing diagram for the circuit, starting from state $(C, J, K, Q) = (0, 1, 1, 0)$, when C changes to 1.

What is the maximum time for which C must remain 1 for the circuit to enter the stable state $(C, J, K, Q) = (1, 1, 0, 1)$?

✓ Answer

Time $\times\tau$	Action
0	$C \to 1$
1	$NAND_1 \to 0$
2	$Q \to 1$
3	$\bar{Q} \to 0$, $NAND_1 \to 1$, $NAND_2 \to 0$

Clock need to be high for less than 3τ.

4

Assume that the circuit P4 operates in fundamental mode and nodes (S, R) are never equal to $(1, 1)$.

a

Determine the transition table for the portion of the circuit comprising gates 1 through 6. Use y5 and y6 as the internal variables.

✓ Answer

They essentially act as latches for S and R, when clock is high, otherwise, keeps the outputs the same

b

What does the entire circuit do? Is it an edge-triggered flip-flop?

✓ Answer

Yes, the first part latches, then the second part flip-flops on that latch that was captured when clock is high

5

Find a state table with a minimum number of internal states that specifies the same external behavior for each of the following state tables.

a

✓ Answer

Kite Table

	1	2	3-1	4-1	5	6-2	7	8
1	o	x	o	o	x	x	x	x
2	-	o	x	x	x	o	x	x
3	-	-	o	o	x	x	x	x
4	-	-	-	o	x	x	x	x
5	-	-	-	-	o	x	x	x
6	-	-	-	-	-	o	x	x
7	-	-	-	-	-	-	o	x
8	-	-	-	-	-	-	-	o

Z

s\ X_1X_2	00	01	11	10
1	0	1	1	0
2	1	0	1	1
3	1	1	0	0
4	1	0	1	1
5	1	0	1	1

S

s\ X_1X_2	00	01	11	10
1	2	3	1	1
2	1	5	1	3
3	2	3	1	4
4	1	2	1	3
5	1	2	1	1

b

✓ **Answer**

Kite Table

	A	B	C	D-C	E	F-E	G-E	H
A	o	x	x	x	x	x	x	x
B	-	o	x	x	x	x	x	x
C	-	-	o	o	x	x	x	x
D	-	-	-	o	x	x	x	x
E	-	-	-	-	o	o	o	x
F	-	-	-	-	-	o	o	x

	A	B	C	D-C	E	F-E	G-E	H
G	-	-	-	-	-	-	o	x
H	-	-	-	-	-	-	-	o

S,Z

s\x	0	1	z
A	B	E	1
B	D	C	1
C	C	D	0
D	C	C	1
E	C	A	0

c

✓ Answer

Kite Table

	A	B	C	D	E-C	F-A	G	H-B
A	o	x	x	x	x	o	x	x
B	-	o	x	x	x	x	x	o
C	-	-	o	x	o	x	x	x
D	-	-	-	o	x	x	x	x
E	-	-	-	-	o	x	x	x
F	-	-	-	-	-	o	x	x
G	-	-	-	-	-	-	o	x
H	-	-	-	-	-	-	-	o

Transition and Z

s	v	w	X	Z
A	D	D	A	0
B	C	D	A	1
C	C	A	A	1
D	D	B	C	0
E	E	E	A	0

6

machine with one level input x and one level output Z. For each machine determine a flow table, with an explanation of the purpose of each state. You may first use a state diagram.

a

$z = 1$ whenever the last five input bits contains exactly three 1's and the string starts with two 1's, After each string that starts with two 1's, the analysis of the next string does not start until the end of the current five bits, whether it produces a 1 output or not. For example, on input sequence 11011010, the output is 00000000, whereas on input sequence 10011010 the output sequence is 00000001.

✓ Answer

Next	**0**	**1**	**Z**
Wait	Wait	1	0
Wait-1	Wait	Wait	0
1	Wait	11	0
11	110	111	0
110	1100	3	0
111	3	Wait-1	0
3	Complete	Wait	0
1100	Wait	Complete	0
Complete	Wait	Wait	1

b

$z = 1$ on every occurrence of an input 1 following two or three consecutive input 0's. At all other times $z = 0$.

✓ Answer

Next	**0**	**1**	**Z**
0	1	0	0
1	2	0	0
2	2	Complete	0
Complete	1	0	1

c

$z = 1$ if the last five input bits are 01101. The machine detects overlapping patterns. For example, on input sequence 01101101, the output sequence is 00001001.

✓ Answer

Next	**0**	**1**	**Z**
Wait	0	Wait	0
0	0	01	0

Next	0	1	Z
01	0	011	0
011	0110	Wait	0
0110	0	01101	0
Complete	0	011	1

7

A circuit is to be designed having two pulse inputs X_1, and X_2 and one dc output z. Whenever an X_1 , pulse is received, the output is to become equal to 1, provided that there been exactly two X_2 pulses after the last previous X_1 pulse. Otherwise, the output is to remain equal to 0. Once the output becomes equal to 1, it is to remain equal to 1 until the next X_1 pulse. Whenever an X_2 pulse is received, the output is to become equal to 0. Write a (pulse-mode) flow table for this circuit.

✓ **Answer**

Next	X_1	X_2	Z
Wait	Ready	Wait	0
Ready	Ready	1	0
1	Ready	2	0
2	Complete	Wait	0
Complete	Ready	1	1

8

Each of the following is a specification for a fundamental mode sequential circuit, for each case find a primitive flow table. Annotate each state with comments,

a

The inputs are w and x; the output is z. $z = 1$ if and only if both w and x have value 1, but only if w became 1 before x became 1. A sample timing diagram is shown in P1.

✓ **Answer**

s\wx	00	01	11	10	Z
Wait	Wait	x	x	w	0
x	Wait	x	False	w	0
w	Wait	x	True	w	0
False	Wait	x	False	w	0
True	Wait	x	True	w	1

b

The inputs are c (clock) and x; the output is z. Output z becomes 1 at the falling edge of the first clock pulse during which x is 1 for the entire clock pulse. z remains 1 until the falling edge of the first clock pulse during which x is 0 for the entire clock pulse. At this time z returns to 0, and the circuit reinitializes itself. A sample timing diagram is shown in P2.

✓ Answer

s\cx	00	01	10	11	Z
000	000	001	010	011	0
001	000	001	011	011	0
010	000	001	010	010	0
011	100	101	010	011	0
100	100	101	110	111	1
101	100	101	110	111	1
110	000	001	110	111	1
111	100	101	110	111	1

c

A D flip-flop with data lockout. The D input is sampled on the rising edge of the clock pulse, and that value becomes the output on the falling edge of the same clock pulse. A sample timing diagram is shown in P3.

✓ Answer

Q\cd	00	01	10	11	Z
000	000	001	010	011	0
001	000	001	010	011	0
010	000	001	010	010	0
011	100	101	011	011	0
100	100	101	110	111	1
101	100	101	110	111	1
110	000	001	110	110	1
111	100	101	111	111	1